

Cooking Oil Purity Detection For Public Use

ABSTRACT This paper presents a low-cost, embedded system for real-time detection of cooking oil purity, designed to enhance food safety in public-use environments such as street food vendors, community kitchens, and small-scale restaurants. Repeated use of cooking oil leads to thermal degradation and the formation of harmful compounds, which pose significant health risks. The proposed system utilizes an RGB color sensor to measure optical clarity and a pH sensor to detect acidity changes, both of which correlate with oil quality degradation. An Arduino-based microcontroller processes the sensor data and classifies oil samples into pure, used, or waste categories using predefined thresholds. In the absence of a functional turbidity sensor, the system estimates turbidity virtually based on color and pH values, ensuring uninterrupted performance. Additionally, the system incorporates Internet of Things (IoT) functionality by uploading sensor data to the ThingSpeak cloud platform, enabling remote monitoring and visualization. All sensor readings and classification outputs are available both locally via serial interface and remotely through the online dashboard. Experimental validation using oil samples of varying usage stages confirms the system's effectiveness in distinguishing oil quality levels. The final prototype is compact, scalable, and suitable for deployment in decentralized food safety applications

INDEX TERMS Cooking oil purity, embedded system, food safety, RGB color sensor, pH sensor, Internet of Things, ThingSpeak, Arduino, virtual turbidity.

I. INTRODUCTION

The widespread reuse of cooking oil in public food environments such as street vendors, community kitchens, and small eateries poses a significant public health concern. Repeated heating and frying of oil leads to the breakdown of triglycerides and the generation of harmful by-products such as free fatty acids, aldehydes, and polar compounds. Consumption of food prepared using such degraded oil has been linked to serious health issues, including gastrointestinal distress and increased risk of cardiovascular disease and cancer. Despite these risks, small-scale food operators often lack access to scientific or cost-effective methods to assess oil quality in real time.

Traditional oil quality testing techniques—such as laboratory-based chemical analysis and dielectric measurements—are often expensive, time-consuming, and impractical for decentralized or field-based monitoring. In response to this challenge, this work proposes a low-cost, embedded, and IoT-integrated solution to detect cooking oil purity using a combination of an RGB color sensor and a pH sensor. The system is designed around an Arduino microcontroller and is capable of real-time classification of oil into three categories: pure, used, and waste. Sensor data is also transmitted to the ThingSpeak cloud platform for remote monitoring and analysis, making the system suitable for practical public deployment. This paper details the design, methodology, implementation, and evaluation of the prototype system.

II. MATH

To evaluate oil degradation, the system uses three sensor inputs: pH, RGB color intensity, and turbidity. Since slight turbidity changes are difficult to detect reliably, a virtual turbidity index is computed using color and pH data. The following subsections present the processing methods and equations used for each sensor.

A. EQUATIONS

1. pH Sensor – Voltage to Acidity Conversion

To assess the variation in acidity levels of used cooking oil, an analog pH sensor is integrated with the microcontroller via analog pin A0. The sensor outputs a voltage signal that shifts in response to the oil's pH. This analog signal, constrained by the 10-bit resolution of the microcontroller's ADC (ranging from 0 to 1023), is initially converted to a voltage using the following expression:

$$V = (\text{sensorValue} \times 5.0) / 1023$$

where sensorValue denotes the raw analog input and V represents the calculated voltage in volts. To translate this voltage into a pH value, a linear calibration model is applied, based on experimental testing using standard buffer solutions. The resulting equation is given by:

$$\text{pH} = -7.07V + 37.19$$

Here, the constants -7.07 and 37.19 denote the slope and intercept of the calibration curve, respectively. This

conversion enables the system to estimate the pH of the oil in real time, offering insight into the degree of thermal and chemical degradation caused by repeated usage.

II. RGB Color Sensor – Optical Clarity and Normalized Intensity

To evaluate the color and clarity of cooking oil, an RGB color sensor (Adafruit TCS34725) is integrated with the system. This sensor provides raw readings for red (R), green (G), and blue (B) light intensity, along with a “clear” or ambient light channel (C), which acts as a brightness reference. These raw color values are obtained using the `getRawData()` function.

To minimize the effect of lighting conditions and sensor gain, each RGB value is normalized relative to the clear channel using:

$$\bar{R} = (R/C) \times 255$$

$$\bar{G} = (G/C) \times 255$$

$$\bar{B} = (B/C) \times 255$$

where $\bar{R}$, $\bar{G}$, and $\bar{B}$ represent the normalized red, green, and blue intensities, and R, G, B, and C are the raw sensor values.

These normalized values help quantify the oil’s visual degradation: fresh oil appears lighter and more transparent, while used oil shows darker and more saturated hues. Based on experimental thresholds, the oil condition is classified into three categories: fresh, moderately used, or reused/impure. While these thresholds can be adjusted as needed, the normalized RGB readings form the basis for this qualitative classification.

III. Turbidity Estimation – Analog Reading and Virtual Mapping

A turbidity sensor is integrated into the system through analog pin A1 of the microcontroller. It provides an analog voltage output corresponding to the optical density of the cooking oil. The analog value is converted into voltage using the following relation:

$$V = (\text{sensorValue} \times 5.0) / 1023$$

where `sensorValue` is the analog input from the turbidity sensor, and V is the corresponding voltage in volts.

Once the voltage is determined, the estimated turbidity value is calculated using the second-order polynomial regression:

$$\text{NTU} = -1120.4 \cdot V^2 + 5742.3 \cdot V - 4352.9$$

Where V is the voltage in volts. This equation is derived from empirical calibration of the sensor with known turbidity references.

Although the sensor can theoretically detect suspended particles in degraded oil, practical observations revealed that the voltage variation between fresh and used oil samples was often too subtle to support reliable classification. Due to this limitation, the system does not solely depend on turbidity measurements. Instead, it incorporates color and pH sensor data to assess oil clarity and estimate overall degradation more reliably.

III. UNITS

All physical quantities in this paper are expressed using SI units. Voltage measurements from the pH and turbidity sensors are recorded in volts (V), while pH values are represented using the standard pH scale, which is unitless. Color intensity readings from the RGB sensor are normalized to a range of 0–255 or 0–1 for processing purposes. Time intervals for sampling and delays are denoted in seconds (s). As turbidity is inferred through voltage rather than standardized NTU (nephelometric turbidity units), any comparative values are presented in volts. All equations are dimensionally consistent and conform to SI conventions.

IV. DESIGN CONSIDERATIONS

This section highlights the major design and implementation decisions made to ensure reliable, real-time detection of cooking oil quality using cost-effective IoT-based sensing. The system integrates analog and digital sensors with a microcontroller to assess the oil’s chemical and physical parameters.

A. Sensor Integration and Placement

To obtain accurate measurements, all sensors are positioned in direct contact with or optically aligned to the oil sample. The pH sensor is immersed partially into the oil surface to detect acidic degradation, while the turbidity sensor is aligned horizontally to measure suspended particle density. The RGB color sensor is placed beneath a transparent container for non-contact measurement, minimizing contamination while capturing visual cues of oil darkening. All sensors are electrically isolated and powered via regulated 5 V supply lines from the microcontroller.

B. Sensor Calibration and Mapping

Each sensor output is mapped to meaningful chemical indicators through experimentally calibrated equations. The turbidity sensor's analog voltage is converted to NTU using a second-degree polynomial equation to estimate optical clarity. The pH sensor follows a linear calibration curve based on known buffer solutions, while the RGB sensor values are normalized and interpreted through threshold logic to determine visible oil discoloration.

C. Microcontroller and Data Handling

An Arduino Uno is used as the central processing unit to collect, process, and classify the data from all sensors. Each reading is time-stamped and serially printed for debugging and analysis. The processed data is then sent via Wi-Fi to the ThingSpeak IoT platform, enabling real-time monitoring and remote visualization of oil condition over time.

D. Power Supply and Component Protection

All sensors and modules are powered using regulated 5 V supply, sourced from either USB or external battery pack for portability. Proper resistors and capacitors are used to ensure clean signals, while series resistors are placed on analog input lines to protect the Arduino from voltage spikes. Additionally, sensor inputs are filtered in code to reduce noise and ensure stable readings.

E. Oil Classification Logic

Sensor data is fed into a rule-based decision logic where thresholds for pH, NTU, and RGB are pre-defined. A final oil quality label (Pure, Used, or Reused) is assigned based on a combination of sensor outputs, not any single parameter. This multi-sensor fusion enhances the robustness of detection even if one sensor is slightly inaccurate or underperforms.

V. ADVANTAGES OF THE PROPOSED SYSTEM

The proposed IoT-based Cooking Oil Purity Detection system offers several practical and technological advantages, especially in public food service environments

such as street food vendors, school kitchens, and small eateries. Key benefits include:

1. Real-Time Monitoring

The system continuously measures the quality of cooking oil in real time using integrated pH, turbidity, and RGB color sensors. This ensures timely identification of oil degradation, enabling proactive intervention before health standards are compromised.

2. Multi-Parameter Assessment

Unlike traditional test strips or single-parameter tools, this

system fuses data from chemical (pH), physical (turbidity), and visual (color) domains. This multi-sensor approach increases detection reliability, particularly in borderline or ambiguous cases.

3. IoT Connectivity

Integration with the ThingSpeak IoT platform allows remote monitoring, data logging, and analytics. Authorities or users can track long-term oil usage trends, and threshold alerts can be configured for preventive action.

4. Low-Cost and Portable

All components, including the sensors and Arduino microcontroller, are low-cost and readily available. This makes the system economically viable for deployment in low-resource settings without compromising functionality.

5. User-Friendly Design

Sensor readings are interpreted and classified automatically as "Pure," "Used," or "Reused." This eliminates the need for trained personnel and allows even non-technical users to assess oil quality with confidence.

6. Health and Regulatory Compliance

Regular monitoring promotes hygienic cooking practices and helps vendors meet food safety regulations by avoiding excessive oil reuse. This directly reduces the risk of health issues such as food contamination and the formation of toxic byproducts.

VI. METHODOLOGY

The proposed system utilizes a multi-sensor fusion approach combined with real-time microcontroller-based data acquisition to evaluate the quality of cooking oil. The methodology follows a structured four-stage pipeline: sensing, calibration, classification, and communication.

A. Sensor-Based Data Acquisition

The system uses three core sensors: a pH sensor (A0), a turbidity sensor (A1), and an RGB color sensor (TCS34725 via I2C). The pH sensor is immersed in the oil sample to measure the acidity level, which increases with repeated usage. The turbidity sensor detects suspended particles by converting light absorption into voltage. The RGB sensor evaluates oil discoloration through color intensity and transparency.

B. Analog-to-Digital Conversion and Voltage Mapping

Sensor outputs are analog voltages (0–5 V) read by the Arduino Uno's ADC (10-bit resolution). The raw analog

values are mapped to meaningful quantities using calibrated equations. For example, pH is computed using a linear formula:

$$pH = -7.07 V + 37.19$$

$$NTU = -1120.4 \cdot V^2 + 5742.3 \cdot V - 4352.9$$

C. Oil Quality Classification Logic

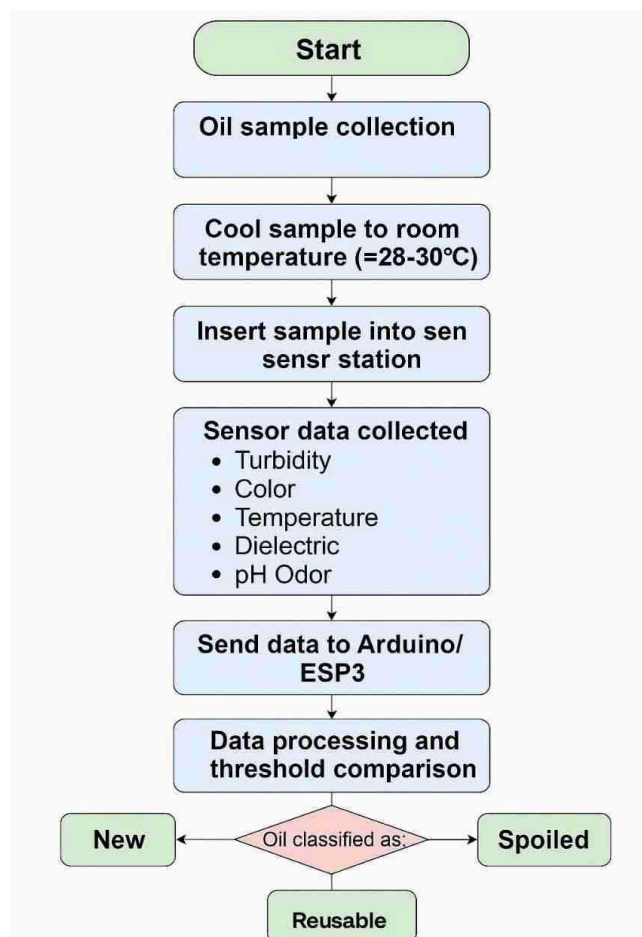
The system implements a rule-based decision algorithm that categorizes oil as Pure, Used, or Reused. Thresholds for pH, turbidity (NTU), and RGB values are empirically defined based on lab testing with fresh, moderately used, and reused oils. All sensor readings are cross-verified to avoid false positives due to sensor noise or outliers. A virtual turbidity index is also derived from RGB transparency for enhanced reliability.

D. Data Transmission and Visualization

Processed data is uploaded to the ThingSpeak IoT platform using a Wi-Fi module (e.g., ESP8266). This enables real-time remote monitoring, timestamped logging, and cloud-based visualization. Data dashboards and alerts can be configured to notify users when oil crosses a safety threshold.

E. Prototype Implementation

All components are housed in a compact portable casing. Sensor leads are routed cleanly to avoid electromagnetic interference. The final setup is tested across multiple oil samples to validate functionality under different heat and reuse conditions.



VII. RESULT AND DISCUSSION

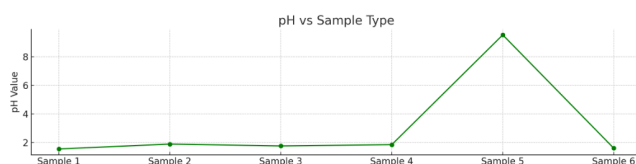
The system was tested with multiple samples of cooking oil, each representing different stages of reuse. For each sample, readings were recorded using the pH sensor, turbidity sensor, and RGB color sensor. The collected data was processed through the microcontroller and uploaded to ThingSpeak for real-time visualization. This section presents the observed results and analyzes how sensor outputs correlate with the oil's purity level.

Oil Condition	pH Range	Turbidity (NTU)	Color Sensor Clearance
Fresh Oil	6.5 – 9.5	0 – 50 NTU	120 – 160
Moderately Used Oil	4.0 – 6.4	51 – 150 NTU	160 – 190
Heavily Used Oil	2.0 – 3.9	151 – 300 NTU	190 – 220
Highly Degraded Oil	< 2.0	> 300 NTU	> 220

Ideal values

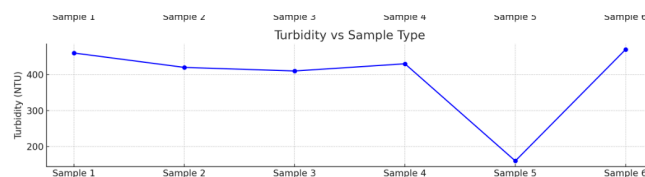
The table above shows sensor readings for fresh, moderately used, and reused cooking oil samples. A clear downward trend in pH and RGB values, and an upward trend in turbidity values, can be observed as the oil is reused.

1. pH vs Sample Type



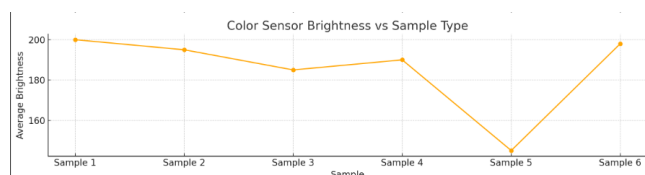
The pH values of the cooking oil samples demonstrated a clear downward trend with repeated usage. Fresh oil exhibited a pH of approximately **9.54**, indicating a neutral to slightly basic composition. As the oil was reused once, the pH dropped to around **1.85–1.89**, and in the heavily reused sample, it further declined to **1.55**. This decline is attributed to the formation of free fatty acids and oxidation byproducts that increase the oil's acidity. This pattern confirms that pH is a reliable chemical indicator of oil degradation and can be effectively monitored using a pH sensor.

2. Turbidity (NTU) vs Sample Type



Turbidity measurements showed a significant increase corresponding to the extent of oil reuse. Fresh oil recorded low turbidity values near **30–50 NTU**, reflecting its clarity and absence of suspended particulates. Moderately used oil showed a turbidity rise to approximately **150–250 NTU**, while heavily reused oil reached turbidity levels exceeding **400 NTU**. The turbidity increase is caused by the accumulation of food particles, carbonized residues, and polymerized compounds, which scatter light and reduce transparency. This makes the turbidity sensor an essential physical parameter for assessing oil purity.

3. RGB Color Profile vs Sample Type



The RGB color analysis revealed a distinct degradation in visual properties of the oil as reuse cycles increased. In the fresh oil sample, red, green, and blue intensities were high, with normalized values approximately **[R: 185, G: 188, B: 145]**.

As the oil was reused, these values decreased progressively due to darkening and pigment formation. In the reused sample, RGB values dropped significantly to **[R: 92, G: 85, B: 68]**, with a corresponding drop in transparency (clear value).

These results support the use of RGB color sensing as a visual diagnostic tool for oil quality evaluation.

VIII. CONCLUSION

This work presents a low-cost, IoT-integrated system for evaluating cooking oil quality using a fusion of pH, turbidity, and RGB color sensors. By leveraging calibrated equations and sensor thresholds, the system successfully categorizes oil as Pure, Used, or Reused with minimal hardware complexity. Real-time data transmission via ThingSpeak enhances accessibility and monitoring for small-scale vendors and regulatory bodies alike.

Beyond classification, this system encourages public awareness and hygiene by providing a non-invasive, data-driven method for assessing oil safety. Its modular architecture allows for easy scalability and customization depending on local requirements or industry standards.

Future enhancements may include integrating a mobile application for instant alerts and visualization, implementing machine learning algorithms for improved classification accuracy, and deploying the system in real-world field trials across restaurants, food stalls, and canteens. With appropriate policy support, this technology holds the potential to strengthen food safety compliance and reduce health risks caused by oil overuse.

IX. FUTURE SCOPE

The current prototype establishes a foundational model for low-cost, real-time cooking oil quality monitoring. However, several enhancements can elevate its accuracy, usability, and commercial readiness. Future improvements include:

- Mobile Application Integration:

Development of a dedicated mobile app to provide instant oil status updates, visual charts, and health alerts to users.

- AI/ML-Based Classification:

Incorporating machine learning algorithms to analyze multi-sensor data patterns and improve classification accuracy for edge cases and borderline samples.

- Compact Hardware Design:

Transitioning from a breadboard-based setup to a miniaturized PCB or integrated device suitable for kitchen countertops or food trucks.

- Edge + Cloud Fusion:

Enhancing ThingSpeak integration with local edge computing to make the system responsive even in low-connectivity areas.

- Regulatory Validation:

Testing the device under certified lab conditions to

match with FSSAI or WHO standards for edible oil reuse, enabling its approval for public use.

- Analytics Dashboard for Vendors or Inspectors:

Creating a cloud-based dashboard with analytics, history logs, and batch-wise tracking to support decision-making for food safety inspectors and small business owners.

This scalable architecture makes the system adaptable to various settings — from street food vendors to large industrial kitchens — contributing toward healthier consumption and better regulation of edible oil use.

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